MPPC (S13371-6050CQ-02)

Jul. 30, 2019

SPECIFICATION SHEET

■ Structure

Parameters	Value	Unit
Effective photosensitive area	5.95 x 5.85	mm^2
Number of channels	4 (2 x 2)	-
Pixel pitch	50	μm
Number of pixel / channel	13,923	-
Package	Ceramic	-
Window	Quartz	-

■ Absolute Maximum Ratings

Parameters	Symbol	Value	Unit
Operating temperature *1	Topr	up to +60	°C
Storage temperature *1	Topr	up to +80	°C
Soldering condition	Tsol	350°C max., once, 3 sec max.	°C

*1: No condensation

When there is a temperature difference between a product and the surrounding area in high humidity environment, dew condensation may occur on the product surface. Dew condensation on the product may cause deterioration in characteristics and reliability.

■ Electrical and optical characteristics

(Typ. T = 25 deg C, Vr = Vop unless otherwise noted)

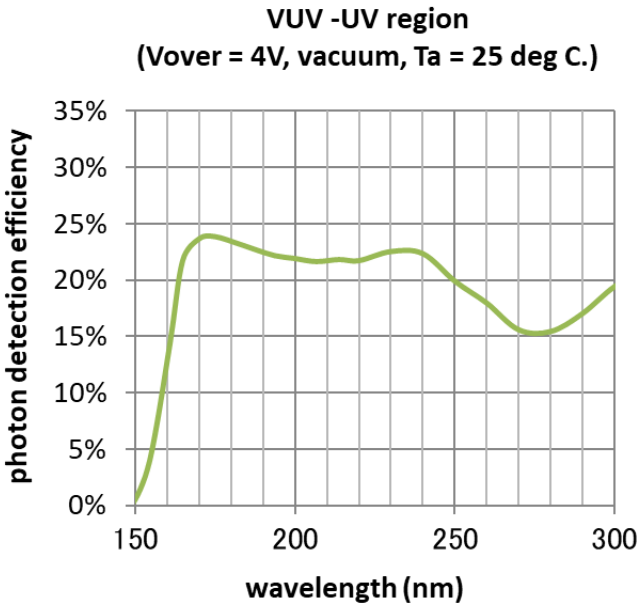
Parameters	Symbol	Value (typ.)	Unit
Spectral response range	λ	155 to 900	nm
Peak sensitivity wavelength	λp	500	nm
Photon detection efficiency at λp *2	-	32	%
Photon detection efficiency at 175nm (in vacuum) *2	-	24	%
Breakdown voltage	V_{br}	53 +/- 5	V
Recommended operating voltage *3	V_{op}	V_{br} + 4.0	V
Dark count rate / channel	DCR	4.0 (MAX:12)	Mcps
Crosstalk probability	P_{CT}	3	%
Terminal capacitance at Vop / channel	Ct	1200	pF
Gain	M	2.55×10^6	-
Temperature coefficient of reverse voltage Around room temperature	ΔTV_{op}	54	mV/deg C

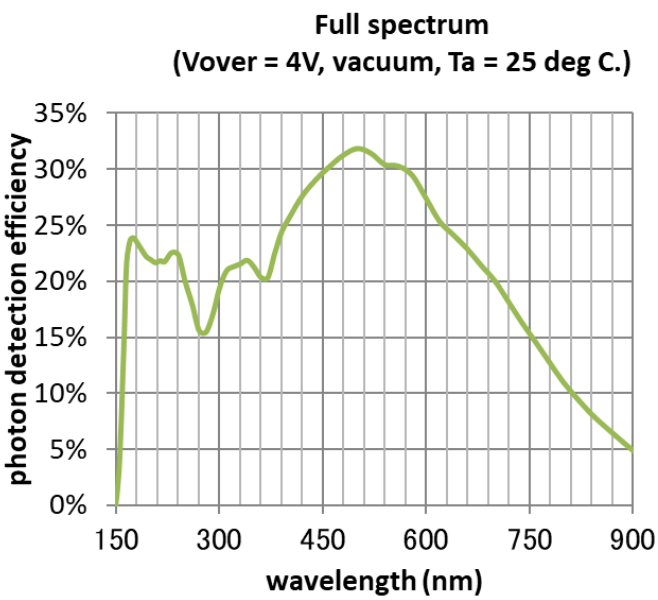
*2 : Photon detection efficiency does not include crosstalk and after pulse.

*3 : Refer to the data attached for each products.

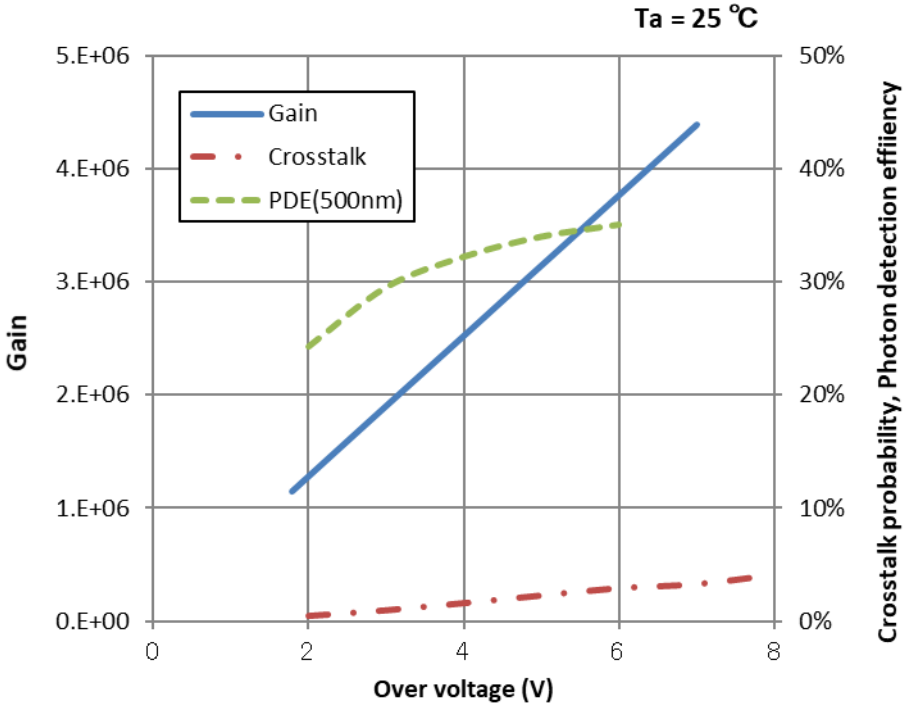
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■ Photon detection efficiency v.s. Wavelength (Vr = V_{op} = V_{br} + 4.0V, measurement example)



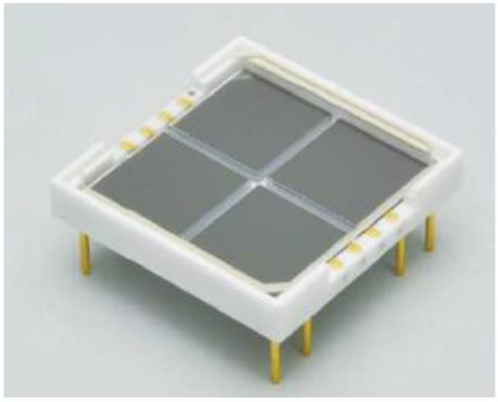


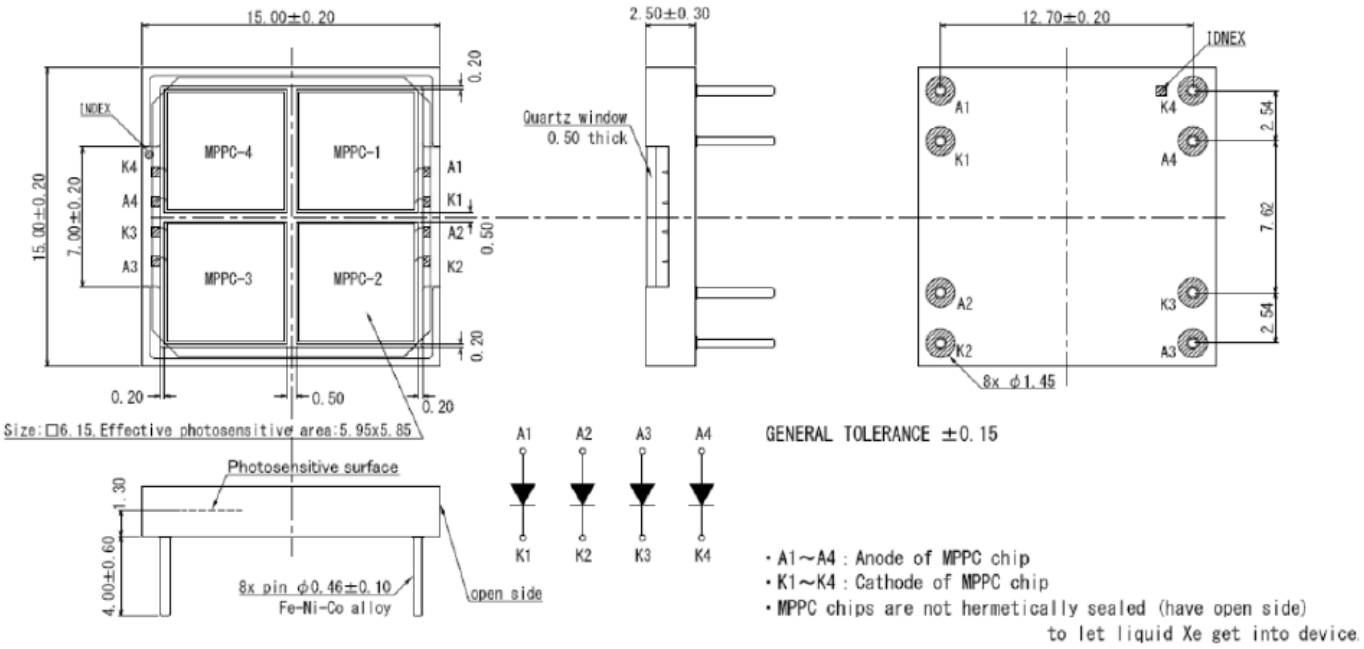
■Overvoltage specifications of gain, crosstalk probability and photon detection efficiency (typical example)



SPECIFICATION SHEET (S13371-6050CQ-02)

■ Dimensional outline (unit: mm)





■ **Precautions**

Please refer below documents for precautions on handling the product.

➢ Precautions for unsealed product

https://www.hamamatsu.com/resources/pdf/ssd/unsealed_KXX-A12014-ENG.pdf